New input file format

b2ag.dat

```
*dimens          (nx, ny, nf, nmax, nx1, ny1; free format)
  128 32  2  2   128 32
*param           (param(0:99); free format)
  -1.0, 0.0, 0.75, 1.0, -0.010, 0.0, 0.040,
  5.0, 1.0, 0.02, 0.05, 0.05, 0.05,
  0.06, 1.0,
  100*0.0
'b2agfs_geometry'     '11276.3500.dpc_.EQI.01.128x32.geo'
```

nx : Parallel dimension of <u>starting</u> mesh

ny : Radial dimension of <u>starting</u> mesh

nf : Oversize factor for matrix allocation

nmax : Maximum number of neighbors in each direction

nx1 : Parallel dimension of <u>basis</u> mesh

ny1 : Radial dimension of <u>basis</u> mesh

b2yt.dat builds a new starting mesh

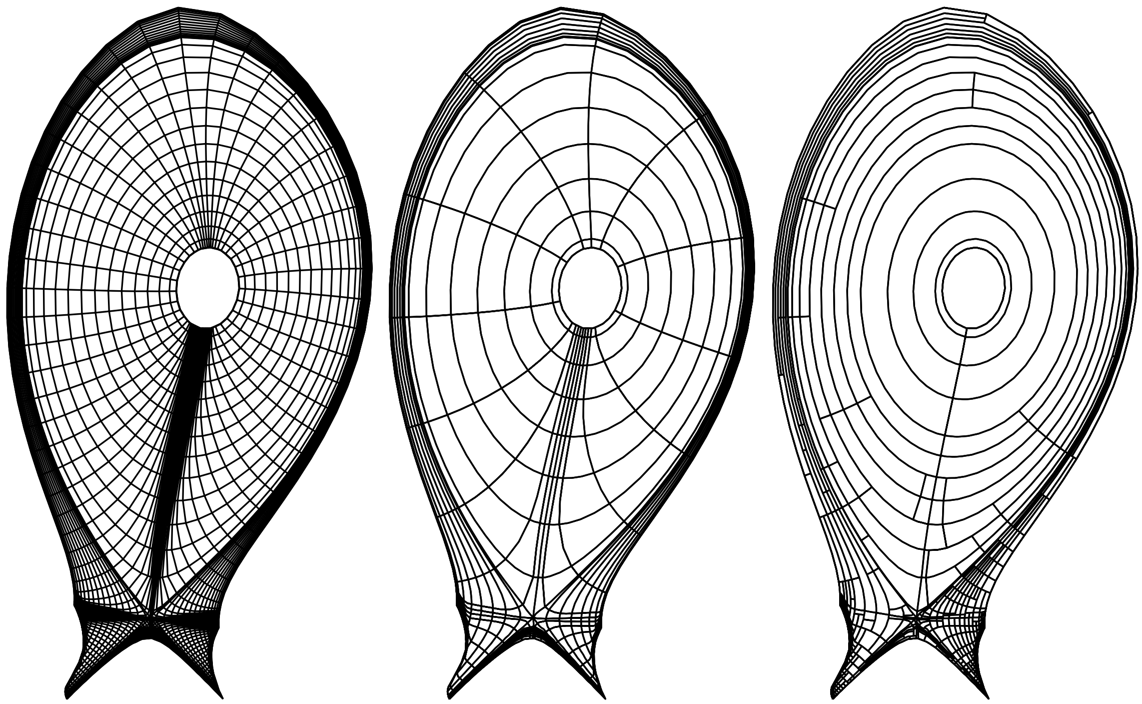
```
*dimens          (nx, ny, nmax; free format)
  96  36  2
```

nx : New parallel dimension of <u>starting</u> mesh

ny : New radial dimension of <u>starting</u> mesh

nmax : New maximum number of neighbors in each direction

Sample grids

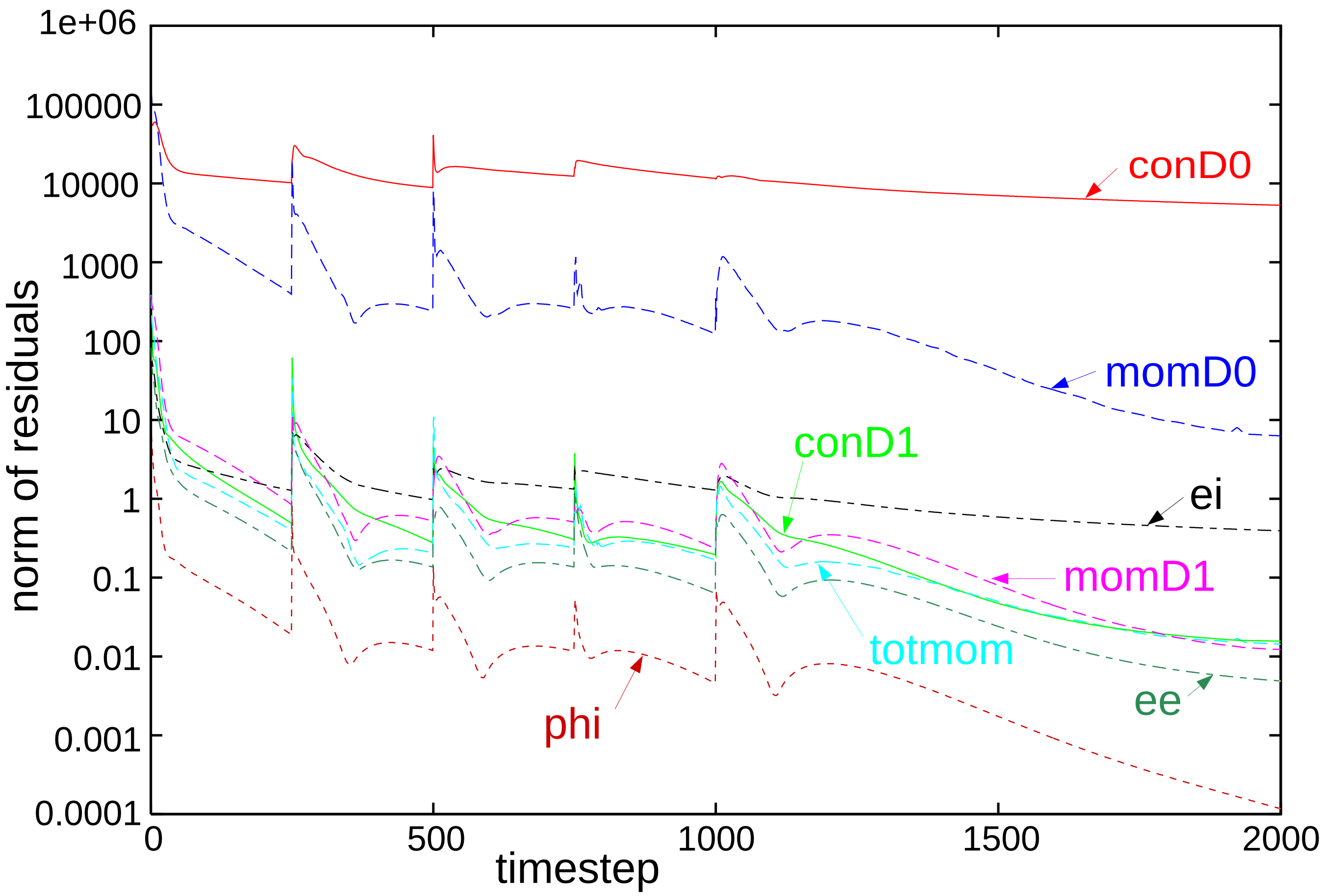


Basis mesh: 128×32 Start mesh: 32×16 Converged state

Very effective grid coarsening in the core region (quasi-1D)

Refinement in the outer divertor

Convergence behaviour



Starting from a converged SOLPS5.0 run

First 250 timesteps Adapting to new interpolation scheme

Every 250 timesteps Grid module called

After a few calls, grid and solution stabilize

Cell and mesh information

Output from the new **kind** command in **b2plot**

Working mesh overlay obtained with command **grid**

Complicated disordered mesh structure

Quick identification of boundaries